Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) with 2+ years building production ML/GenAI infrastructure, distributed deployment automation, and cloud platforms using Java, Python, Go, AWS, and Kubernetes. Current research applies LLM pipelines, RAG, and prompt engineering to agentic Android test automation.

EDUCATION

Master of Science in Computer Science Aug 2024 to Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Database Theory, Intelligent Data Analysis.

Bachelor of Engineering in Computer Engineering Aug 2018 to Jul 2022
AISSMS College of Engineering
Relevant Coursework: Object Oriented Programming, Data Mining, Machine Learning.

TECHNICAL SKILLS

Programming: Java, Python, Go, TypeScript, JavaScript, C/C++, SQL.

ML Lifecycle: MLOps, GenAIOps, Model Training/Validation, RAG, Prompt Engineering, LLM Evaluation.

Data & Databases: PySpark, Delta Lake, PostgreSQL, MySQL, Elasticsearch, Redis, Vector Databases.

Developer Tools: Git, GitLab CI/CD, Jenkins, AWS (EC2, S3, Lambda, SQS), Docker, Kubernetes, ArgoCD, Terraform.

AI & ML: Gemini, Vertex AI, PyTorch, scikit-learn, LangChain, pandas, NumPy, statistical modeling.

WORK EXPERIENCE

Graduate Researcher | University of Cincinnati Feb 2026 to Present

- Designed two stage memory augmented LLM pipeline for Android bug reproduction, decoupling video understanding from device automation across **53 evaluation runs**.
- Benchmarked against GPT-4o + GroundingDINO + CLIP baseline, cutting latency **25.7x** (39.4s vs 1012.1s) while preserving task fidelity across **20+ apps**.

Software Development Engineer | Siemens Industry Software Jul 2022 to Jun 2024

- Automated **Teamcenter X** infrastructure workflows with Python and Boto3, cutting deployment time **40%** across AWS environments.
- Executed **44 production upgrades** in 4 months, debugging Linux, EC2, database, and service issues in live environments.
- Designed Ansible Blue Green deployment framework with rollback support, reducing production upgrade downtime by **50%.**
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Amazon EFS, and Vault.
- Increased automated code coverage from **35% to 78%** as Quality Coach; mentored **5 engineers** on standards and reviews.

Software Developer Intern | Siemens Industry Software Mar 2022 to Jun 2022

- Automated AWS provisioning with Terraform and Go Terratest, cutting deployment time **30%** and catching **95%** misconfigurations.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 to Dec 2021

- Built React and MySQL workflow app digitizing record management and improving departmental operating efficiency by **40%.**

PROJECTS AND RESEARCH

PolicyPilot Medical Benefit Policy Intelligence Portal [React, TS, Node, Gemini, MCP] May 2026

- Architected evidence grounded **RAG pipeline** over Gemini 2.5-flash with citation backed answers, eliminating hallucination in high stakes eligibility decisions.
- Designed **17 tool MCP server** exposing structured LLM tool-calling for PA extraction and cross-payer comparison, consumable by external AI agents.

NYC Mobility Forecasting [PySpark, Delta Lake, scikit-learn, FastAPI] Jul 2026

- Built Spark ML pipeline forecasting demand across **263 zones** from **2.3M** hourly rows; served predictions through a FastAPI model deployment layer.
- Improved MAPE from **18.77% to 16.52%** and cut RMSE **43%** with a two stage Poisson hurdle statistical model over baseline Random Forest.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).